

TASK 1

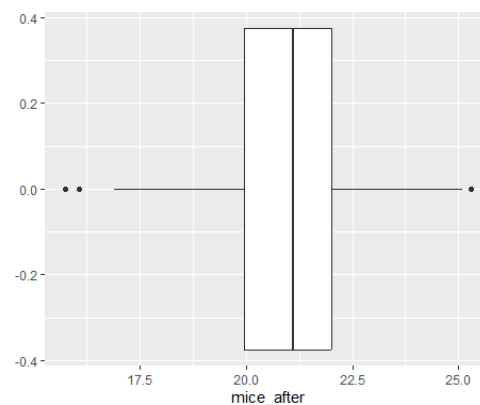
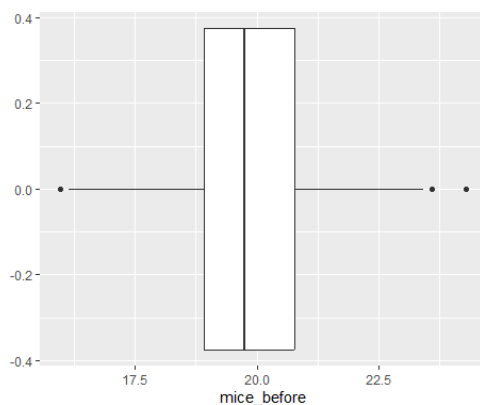
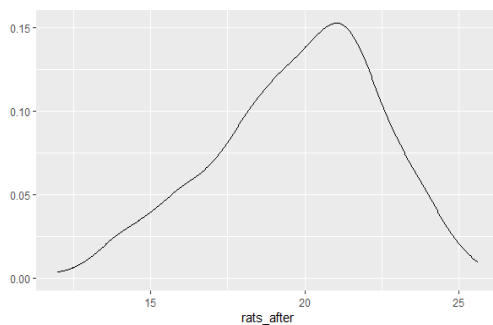
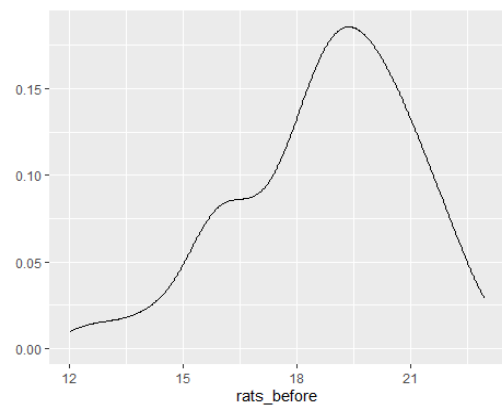
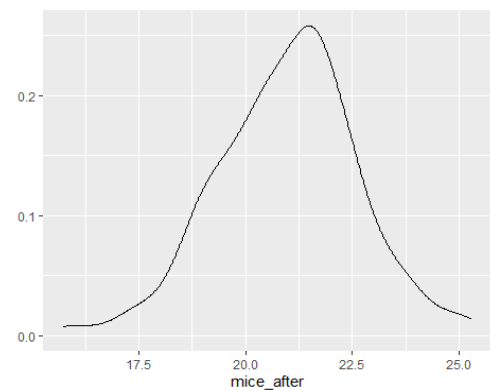
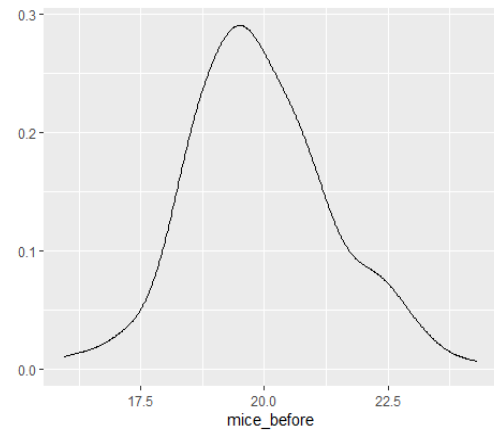
```
rm(list = ls())
set.seed(1234)
before_mice=rnorm(200,mean=20,sd=1.414214)
set.seed(111)
after_mice=rnorm(200,mean=21,sd= 1.581139)
mice=as.data.frame(cbind(before_mice,after_mice))
head(mice)
```

```
set.seed(112)
before_rats=rweibull(200, shape=10, scale = 20)
set.seed(113)
after_rats=rweibull(200, shape=9, scale = 21)
rats=as.data.frame(cbind(before_rats,after_rats))
head(rats)
```

```
par(mfrow=c(1, 2))
qplot(before_mice,data=mice, geom="density")
qplot(after_mice,data=mice, geom="density")
```

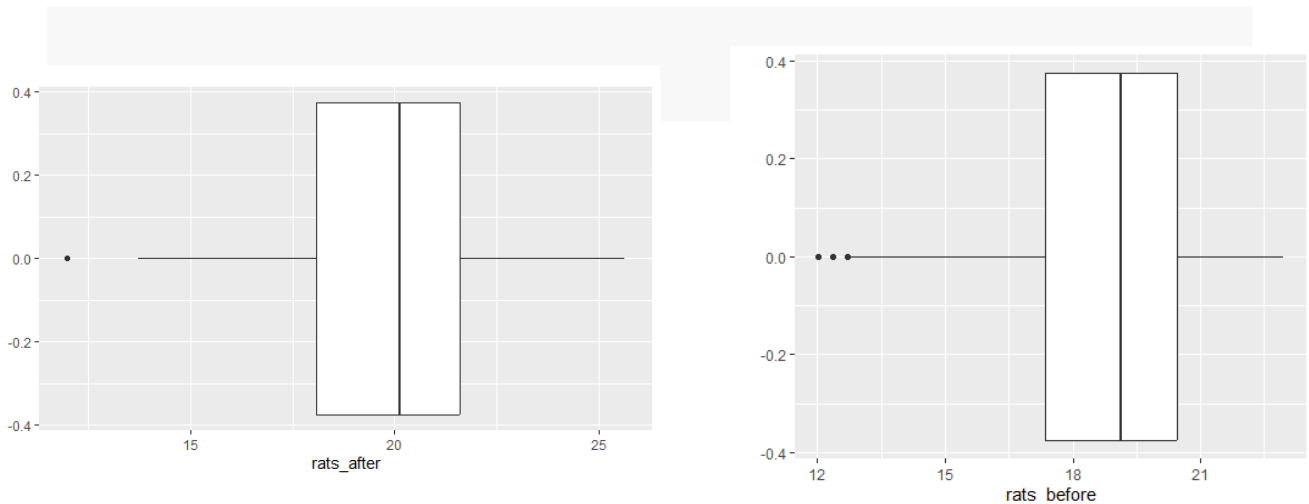
```
par(mfrow=c(1, 2))
qplot(before_rats ,data=rats, geom="density")
qplot(after_rats,data=rats, geom="density")
```

```
par(mfrow=c(1, 2))
qplot(before_mice,data=mice, geom="boxplot")
qplot(after_mice,data=mice, geom="boxplot")
qplot(before_rats,data=rats, geom="boxplot")
qplot(after_rats,data=rats, geom="boxplot")
```



Analysis and results: Similarly, while the boxplot of rats' data appears to be symmetrical, it is not.

qplot(rats_before, rats_after, data = rats, geom = c("boxplot"))



TASK 2

```
par(mfrow=c(1, 2)) qqPlot(mice$before_mice)

## [1] 178 192

qqPlot(mice$after_mice) ## [1] 137 31

shapiro.test(mice$before_mice)

##
## Test for checking normality (Shapiro-Wilk) ##
## mice$before_mice (used a s data)
##
## p-value is 0.007434
## W is 0.98068

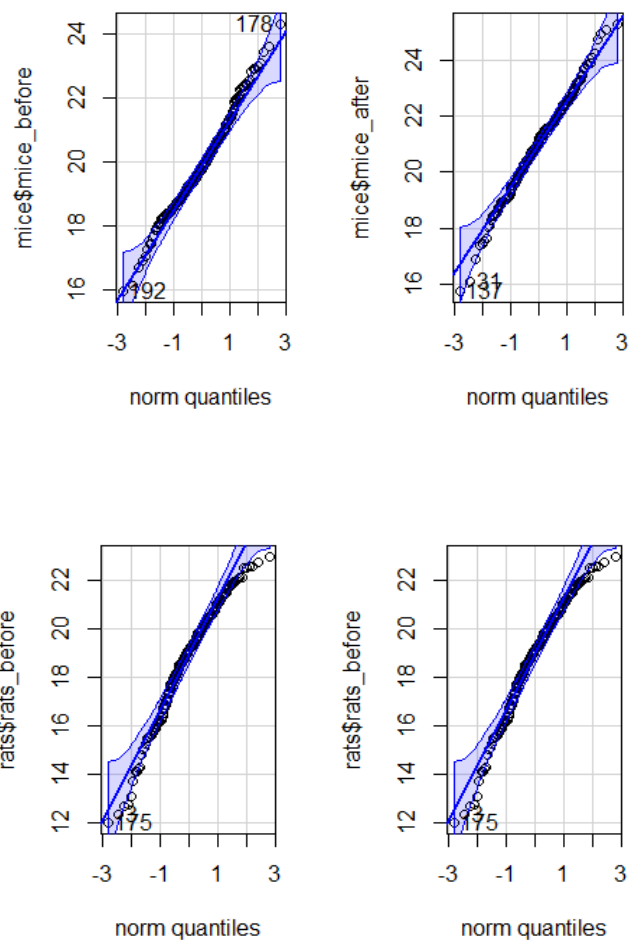
shapiro.test(mice$after_mice) ##
## test for normality ##
## (Shapiro-Wilk) data: mice$after_mice
## p-value = 0.4332, W = 0.99279

par(mfrow=c(1, 2)) qqPlot(rats$before_rats)

## [1] 175 3

qqPlot(rats$before_rats) ## [1] 175 3

shapiro.test(rats$before_rats)
```



Interpretation: Here, qqplot and Shapiro test reveal that the mice before and after the dataset have a normal distribution because the Shapiro test value is greater than 0.05, indicating that the variables are normal. However, qqplot and Shapiro test results suggest that rats before and after both datasets do not follow a normal distribution since the Shapiro test value is less than 0.05, indicating that variables are not normally distributed. So, we'll use the t-test for the mice dataset since it's a parametric test, and the Wilcox test, being a non-parametric test, for the first dataset, will be used. (for rats)

TASK 3

Do t-test

```
t1 <- t.test(before_mice, after_mice, paired = TRUE)
print(t1)
```

```
## t-testing in par ##
## Here, we use as data:
## before_mice and after_mice; 1.188e-10( is the p-value)
## -degree of freedom = 199,
## alternate theory: the actual mean difference is a value other than zero
## CI is ninety five percent; 6.8009( the Value of t),
## -1.3650512 -0.7513789
```

Interpretation:

t-test -6.8009; Indicates that the difference between the means of the groups (or conditions) is substantial relative to the variability within each group.

Degree of freedom: 199. This indicates that there is enough variability in the data to estimate the parameters of the underlying distribution accurately.

P- Value: 1.188e-10. We would not accept the null hypothesis and say that there is a significant difference or effect in the data if the p-value was that low.

Confidence Interval: 95 Percent. This means approximately 95 percent of these intervals would contain the true population parameter.

95 percent confidence interval:

```
## -1.3650512 -0.7513789
```

Sample Estimates: -1.058215. This suggests that, on average, there is a decrease of approximately 1.058215 units (whatever units are relevant to your study) between the paired measurements or conditions.

Here t-test value is large, so it indicates the rejection of null hypothesis which is both variables means are same. Second total number of observations in dataset is 200 so d.f is equal to n-1, and we get 199. Thirdly, p-value is less than 0.05 which also shows our null hypothesis is rejected and concludes that both groups means are different.

Fourthly, Confidence interval for mean is -1.3650512 to -0.7513789 and lastly mean of the difference of both group is -1.058215.

So, t.test shows mice before after weights are different because p value is less than 0.05 which represents both groups' means are significantly different.

##Alternative test:

```
s1 <- wilcox.test(before_rats, after_rats, paired = TRUE)
print(s1)
##
## correction of continuity test (Wilcoxon signed ranked) ##
## after_rats & before_rats used here as dat ##
# 6640 is V, 3.18e-05 is the p-value
## hypothesis: true location shift is a value other than 0 (alternative)
```

Interpretations:

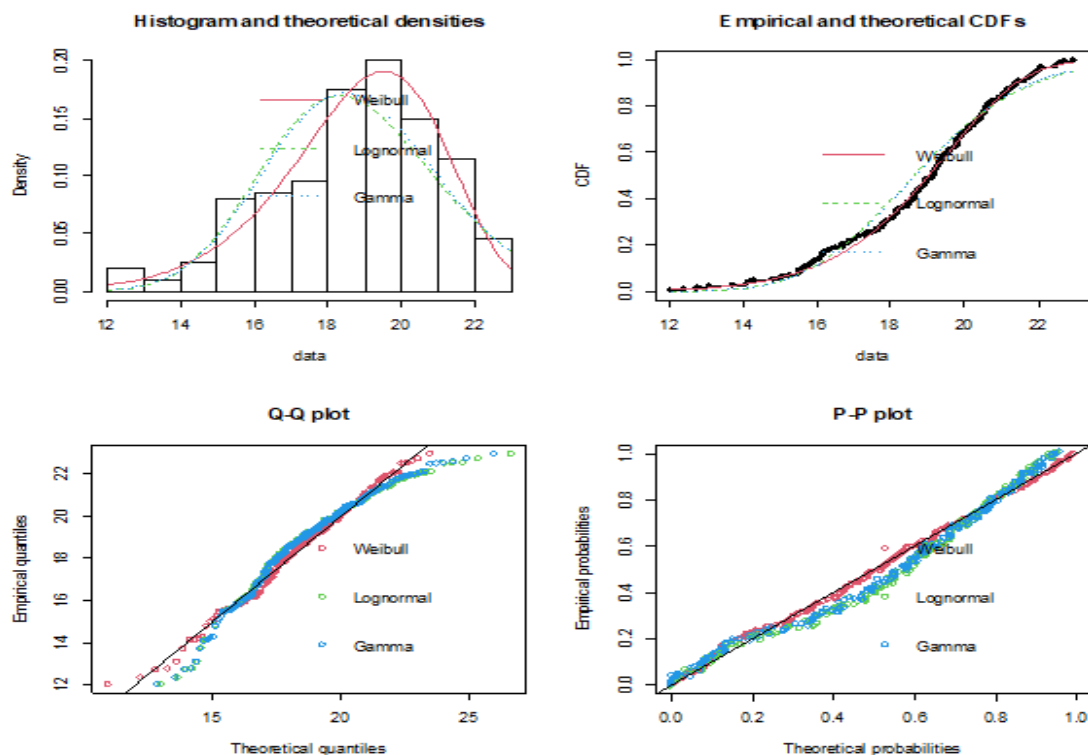
V : 6640; It indicates the overall large magnitude of the discrepancy between the paired measurements

P-value: 3.18e-05; Suggests robust evidence against the null hypothesis, supporting the presence of a significant difference between the paired measurements.

Rats before and after weights are different, according to the Wilcoxon test, since the p value is less than 0.05, indicating that the position shifts of both groups are statistically different.

TASK 4

```
fit_weibull_before <- fitdist(before_rats, "weibull")
fit_lognormal_before <- fitdist(before_rats, "lnorm")
fit_gamma_before <- fitdist(before_rats, "gamma")
fit_weibull_after <- fitdist(after_rats, "weibull")
fit_lognormal_after <- fitdist(after_rats, "lnorm")
fit_gamma_after <- fitdist(after_rats, "gamma")
# Compare fits
par(mfrow = c(2, 2))
denscomp(list(fit_weibull_before, fit_lognormal_before, fit_gamma_before), legendtext = c("Weibull", "Lognormal", "Gamma"))
cdfcomp(list(fit_weibull_before, fit_lognormal_before, fit_gamma_before), legendtext = c("Weibull", "Lognormal", "Gamma"))
qqcomp(list(fit_weibull_before, fit_lognormal_before, fit_gamma_before), legendtext = c("Weibull", "Lognormal", "Gamma"))
ppcomp(list(fit_weibull_before, fit_lognormal_before, fit_gamma_before), legendtext = c("Weibull", "Lognormal", "Gamma"))
```



Interpretation: In the graphs red line is most appropriate for dataset of rats before and that represents Weibull distribution, as it minimizes the difference between the observed data and the theoretical distribution. From the output, The Weibull distribution fits the data well across all plots, closely matching the observed data's shape, cumulative distribution, and quantiles.